Modelling the Habitat Suitability of alpine plant species in the Western Balkans

Labmeeting – 24th of February 2026

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Aims

Motivation and main aims

Based on limited knowledge about the distribution of the focal species, I would like to describe their current potential range, threats in the future & behaviour since the LGM. I would also like to emplhisize weaknesses of this approach.

1. Coarse SDMs with temporal extrapolation
2. Various scale (grain) SDMs with comparision of information loss
3. *Fine models based on remote sensed predictors*

- balkan endemic (sub)alpine flora
- fieldwork done (2020 – 2023)

Aims, elaborated

1. Compose SDMs based on **coarse climate**, **topography** & **geology** data

- evaluation of "simple" models and their ecological relevance (e.g. errors delivered by the climate models, samplesize, effect of the site specific microclimate)
- spatial interpolation
- temopral extrapolation (LGM, future scenarios)

2. Composing SDMs with same data as in **1.**, but downscaling climate models

- comparing finer models with models delivered from **1.**
- estimate information lost in

3. Remote sensing SDMs

Aims, elaborated

- optional
- 20 m grain (1.2 billions cells within AOI) – very high computaional demands
- ecology complicated, almost ecologically inexplainable, predictors indirect
- BUT, predictors “directly” measured on the site and therefore “exact”
- combination with DEM could be powerful

SDM principles

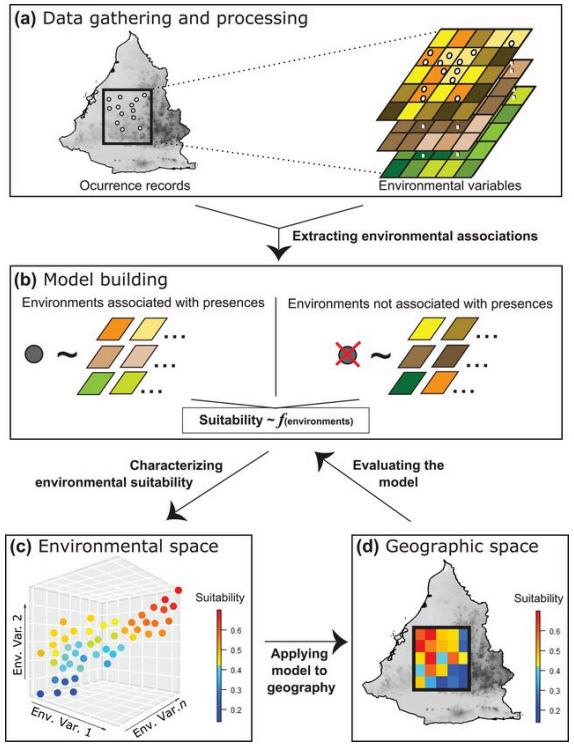


Figure 1: Model fitting and evaluation

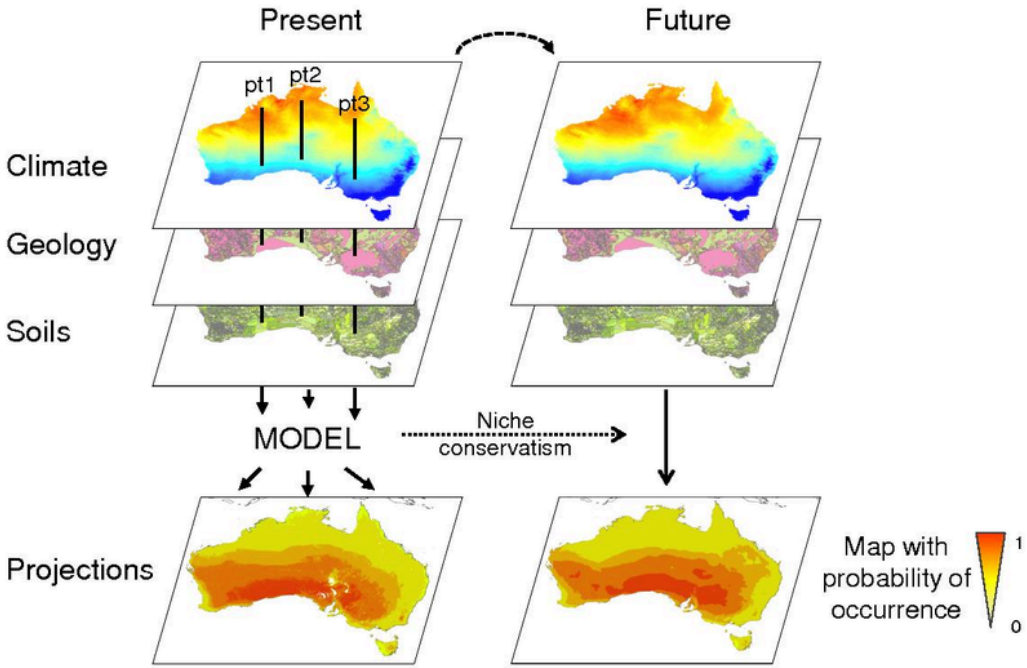


Figure 2: Model projection

Grain question

- one of the key question when

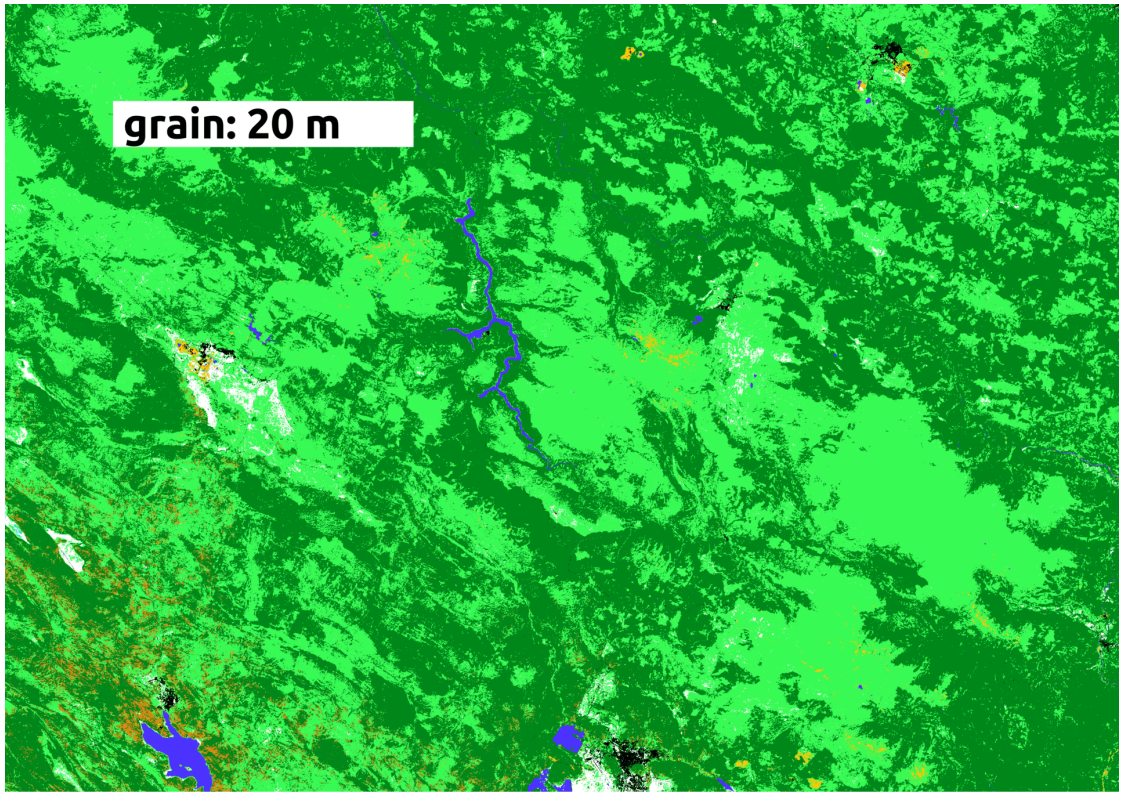


Figure 3: Various grains

- This is a `#blank-slide`.
- Available **themes**[1]:

bluey, reddy, greeny, yelly, purply, dusky, darky.

- Or just use **your own theme color**:
  - `theme: rgb("30500B")`

[1] Use them as **color** functions! e.g., `#reddy("your text")`

Outlined slide

- Check out the **progress bar** at the bottom of the slide.

  `show-progress: true`

- Outline slides with `outlined: true`.

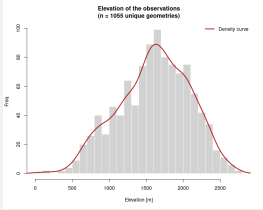
This is a `#grayed` text. Useful for equations.

$$P_t = \alpha - \frac{1}{\sqrt{x} + f(y)}$$

Columns

Columns can be included using `#cols[...][...]`

And this is



an example. Figure 4

- Custom spacing: `#cols(columns: (2fr, 1fr, 2fr), gutter: 2em)[...]`
  - Add a **bibliography slide**...

1. `#let bib = bibliography("you_bibliography_file.bib")`
2. `#bibliography-slide(bib)`